University of Pisa
Department of Physics
Master's Degree in Physics of Complex Systems

Anomalous Transport in Soaring Flights

Collective Strategies and Intermittent Search Models

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Abstract

Soaring birds and human pilots stay aloft by extracting energy from atmospheric updrafts, producing long, intermittent trajectories that alternate climbs within thermals with quasi-ballistic glides. Such motion is a candidate for *anomalous transport*, that is, a departure from ordinary Brownian diffusion. This thesis characterizes the transport statistics of real soaring flights from a large empirical dataset and compares them with stochastic transport models.

This document reports the current state of the project. A first, reproducible acquisition has collected 212,602 flights from the FFVL Coupe Fédérale de Distance across two disciplines: 203,343 paraglider flights (seasons 1999–2000 to 2025–2026) and 9,259 hang-glider flights (2001–2002 to 2025–2026). Of these, 192,971 carry a GPS tracklog and 192,768 have been downloaded and verified. Data collection is ongoing and may be extended to further public soaring databases – notably WeGlide for sailplanes, and XContest – using the same approach. The remainder of the document describes the acquisition method, the dataset and its statistics, and the planned analysis and modelling.

Contents

Chapter 1

Introduction

1.1 Motivation

Gliders, paragliders, and soaring birds carry no continuous source of propulsion; to remain airborne they extract energy from the moving atmosphere, primarily from *thermals*, the buoyant updrafts that form as sunlight heats the ground. A cross-country flight thus proceeds as a repeating cycle of three phases. In the *transition* phase the flyer glides along a nearly straight, energy-efficient path toward the next source of lift. If none is reached at a safe altitude, a *search* phase follows, during which the flyer probes the surrounding air mass to locate lift. Once lift is found, a *climb* phase begins, in which the flyer circles within the rising air to regain altitude. The cycle alternates long, directed displacements with episodes of slow, localized motion.

Intermittent and directionally persistent motion of this kind is the setting in which *anomalous transport* arises. Let $\mathbf{r}(t)$ denote the horizontal position of a flight, measured from take-off. Transport is anomalous when the mean-squared displacement grows as a non-trivial power of time, $\langle|\mathbf{r}(t)|^2\rangle \sim t^\alpha$ with $\alpha \neq 1$, as opposed to the linear growth ($\alpha = 1$) of ordinary Brownian diffusion. Continuous-time random walks and Lévy walks are the standard frameworks for such regimes [1,2], which have been reported in animal movement, in disordered media, and in turbulent flows. Whether the trajectories of real soaring flights belong to an anomalous regime, and of which type, is the question addressed in this thesis.

Soaring has previously been studied mainly as a control problem—how to centre and exploit a thermal efficiently, including by reinforcement learning [3]. The present work is statistical instead: it asks which transport properties emerge at the level of the ensemble of observed trajectories. Its direct precedent is the study of Vilpellet et al. [4], who analysed a large set of human soaring flights, found a horizontal transport that is sub-ballistic and approximately independent of aircraft type (Hurst exponent $H \approx 0.88$), and introduced a phase-resolved analysis based on the transition–search–climb cycle. This thesis builds on that study: it enlarges the dataset and re-examines both the phase segmentation and the transport analysis.

1.2 Objectives

The thesis pursues three objectives:

1. to assemble a large, reproducible, and documented dataset of real soaring flights;

2. to reconstruct clean trajectories from the raw GPS logs and define suitable transport observables;

3. to test for anomalous transport and to compare the empirical statistics with stochastic transport models.

A first dataset—the FFVL CFD archive of paraglider and hang-glider flights—has been assembled and verified and is described here; data collection is ongoing and may incorporate further sources, notably sailplanes (Chapter 2). The second and third objectives define the work ahead (Chapter 4).

1.3 Scope of this document

This document reports the current state of the project and is revised as the work proceeds. Its quantitative content—the summary statistics and tables—is generated directly from the dataset, so it remains consistent with the data in hand. Chapter 2 describes the acquisition of the data and the reproducibility of the pipeline; Chapter 3 describes the dataset and its statistics; and Chapter 4 sets out the planned analysis and modelling.

Chapter 2

Data acquisition

2.1 Source: the FFVL Coupe Fédérale de Distance

The data come from the *Coupe Fédérale de Distance* (CFD), the cross-country competition run by the French free-flight federation (FFVL). The federation operates two parallel competitions on separate sites: one for paragliders (`parapente.ffvl.fr`) and one for hang gliders (`delta.ffvl.fr`). Pilots upload their flights; each entry carries metadata (pilot, date, take-off and landing sites, scored distance, duration, glider) and, in most cases, the raw GPS tracklog in IGC format. Both sites expose identical URL structure and machine-readable XML exports, so the same acquisition pipeline handles both. A "season" is identified by its starting year, e.g. the year 1999 denotes the 1999–2000 season; the paraglider archive starts in 1999–2000 and the hang-glider archive in 2001–2002.

2.2 What is collected

For every season the pipeline retrieves two things:

- the **season XML export**, archived verbatim as provenance (`raw_xml/<year>.xml`); it is the authoritative list of flights and links;
- the **IGC tracklog files** referenced by the XML, one file per flight that has a track.

From these, a derived **catalog** is built (`catalog.csv`, one row per flight, metadata plus the local path of its track) together with a per-season index (`seasons_index.csv`). The raw data (tens of gigabytes) live on an external disk and are never committed to the repository; only the small per-season index is versioned, and it is the source of the statistics in Chapter 3.

2.3 Pipeline and tooling

Acquisition is performed by a dedicated, installable Python package (`soaring.acquisition.ffvl`), exposed through two command-line tools: `soaring-para` for paraglider data and `soaring-delta` for hang-glider data. Both share the same subcommands: `fetch-xml` archives the season XMLs, `download` downloads the IGC files, `build-catalog` rebuilds the catalog and index, `status` reports per-season coverage, and `verify` checks the integrity of the files on disk. The design favours robustness for a long-running, large job:

Resumable downloads. Files already present are skipped, so the job can be interrupted and resumed freely.

Atomic writes. Each file is written to a temporary `.part` and then renamed, so a file under its final name is always complete – important on the exFAT external disk, which has no journaling.

Content validation. A response is accepted only if it looks like a real IGC file (an `A` header record followed by at least one `B` fix record); HTML error pages and truncated responses are rejected and retried.

Courteous networking. A small bounded thread pool, per-request timeouts, exponential backoff, and a minimum delay between requests avoid hammering the server; a browser TLS fingerprint is impersonated to pass the site's Cloudflare challenge.

2.4 Reproducibility

The destination directory (`data_root`) is not hard-coded: it is set per machine through an environment variable (`SOARING_PARA_DATA_ROOT` for paragliders, `SOARING_DELTA_DATA_ROOT` for hang gliders) or a configuration file shipping a placeholder (`configs/para_download.yaml` and `configs/delta_download.yaml`), so no machine-specific path is committed. Every command validates this setting at start-up and aborts with an explicit message if the target is missing, rather than failing obscurely. Archiving the season XMLs verbatim keeps the full provenance: the catalog can always be rebuilt from them with `build-catalog`.

2.5 Integrity

Because the data sit on an external drive, the `verify` subcommand re-scans every downloaded file on disk, reusing the same IGC validation applied at download time, and flags truncated or invalid files, leftover `.part` fragments, and read errors. Both FFVL collections have been verified in full: all 186,052 paraglider and 6,716 hang-glider downloaded tracklogs pass validation (Chapter 3).

2.6 Further data sources

The FFVL CFD is the primary source of the present study, but the dataset is not assumed to be final. The paraglider data (~203 000 flights, ~186 000 tracklogs) are the main analysis target; the hang-glider data (~9 300 flights, ~6 700 tracklogs) are collected in parallel and enable the cross-type comparison of Chapter 3. Additional public soaring repositories are being pursued to enlarge and diversify the sample and, above all, to add the third glider category, *sailplanes* (Section 3.3). The most complete sailplane source is *WeGlide*: it exposes flight metadata and IGC tracks through an API, but the default access is rate-limited (of order sixty requests per day), so a bulk export of the archive is not feasible under it; a request for research access with a higher quota has been sent and is pending. Paraglider and hang-glider coverage may likewise be broadened through further portals such as XContest. These sources expose comparable flight metadata and IGC tracks and can be acquired with the same approach and merged into the same catalog.

Chapter 3

The dataset

3.1 The IGC tracklog format

A flight track is stored as an *IGC file*, the plain-text format defined by the FAI/IGC Technical Specification for IGC-approved GNSS flight recorders [5]; it is the standard output of the recorders used in free-flight competitions. Its layout is therefore fixed by that standard, not inferred from the files, and the acquisition validates every downloaded track against it (Chapter 2), so conformance is checked across the whole archive.

Each line is a *record* whose type is set by its first character – a code assigned by the standard, not chosen by the logger. The records relevant here are:

`A` the flight-recorder/manufacturer identifier (first line of the file);

`H` header records (flight date, pilot, glider, geodetic datum, ...);

`I` the declaration of the optional extra fields appended to every fix;

`B` the GPS fixes – the body of the file.

A `B` record stores its fields at fixed character positions. Decoded against the standard, the example

```
B 094329 4933044N 00005193E A 00095 00149
```

reads as: UTC time `09:43:29`; latitude $49°33.044'$ N and longitude $000°05.193'$ E on the WGS84 datum; a fix-validity flag; and *two* altitudes in metres – the barometric (pressure) altitude (95 m) followed by the GNSS altitude (149 m).

The validity flag is fixed by the standard to two values and certifies the *GNSS altitude* rather than the horizontal position: `A` marks a valid three-dimensional fix (usable GNSS altitude), whereas `V` marks a two-dimensional fix or a GNSS drop-out, in which case the standard requires the GNSS-altitude field to be written as zero. A `V` fix can thus still carry a usable latitude/longitude and a valid barometric altitude.

Decoding a `B` record from fixed character positions can succeed numerically while the result is not a valid encoding at all – a corrupted byte can still parse as a digit. The parser therefore checks, at decode time and independently of any flight-dynamics judgement, that the time is a valid 24-hour `hh:mm:ss`, that the arc-minutes field of latitude and longitude is `<60` (the `DDMM.mmm` encoding is undefined otherwise), that the hemisphere letter is one of the two valid characters, and that the decoded latitude/longitude fall within $[-90, 90]/[-180, 180]$ degrees; a record failing any of these is dropped as unparseable, on the same footing as a record with the wrong length or a non-numeric field; this check is implemented in the parser now and covered by unit tests. It is a *format*-validity check – is the record even a legal encoding – distinct from the *plausibility* checks of fix-level cleaning (Chapter 4, e.g. the inter-fix speed bound), which are designed and thresholded but not yet implemented: they target fixes that decode validly but would be physically implausible given the rest of the trajectory.

The two altitude channels differ in reference and in noise – the barometric value is referenced to the ICAO standard atmosphere and is smooth, the GNSS value is referenced to the WGS84 ellipsoid and is vertically noisier – and on older loggers one channel may be absent (recorded as zero). The analysis adopts the *barometric* channel for the vertical dynamics, for the signal-quality reasons set out and quantified in Chapter 4; this choice matters because the vertical velocity derived from it drives the phase segmentation.

The sequence of `B` records is thus a time-stamped three-dimensional trajectory. The sampling period is typically a few seconds, is not guaranteed to be uniform, and varies across recording devices – a point the analysis must handle explicitly.

3.2 Flight metadata and the catalog

Every flight in the season XML carries a set of metadata as record attributes, which the acquisition preserves. Beyond the identifiers and links, the fields used downstream are: the **date**; the **pilot** and **club**; the **take-off** and **landing** sites and the French **department**; the glider, given as both a model name (`aile`) and a class (`aile_class`, Section 3.3); and the competition quantities – the **flight type** (the *scoring* category: free distance, out-and-return, flat or FAI triangle, hike-and-fly, . . .), the scored **distance** and **points**, and, where present, the **duration** and average **speed**. The flight type is a scoring label and must not be read as a glider or manoeuvre type.

From these, the pipeline builds a single derived table, the **catalog** (`catalog.csv`), with *one row per flight* across all seasons: the metadata above plus a `local_path` linking each flight to its track on disk and a `downloaded` flag. It is the working index of the project – a flight can be traced to its file, and back (through the identifier embedded in the file name) to its metadata and public page, without any external lookup. A smaller **season index** (`seasons_index.csv`), versioned in the repository, summarises each season by the number of flights declared, carrying a track, and downloaded; it is the source of the statistics below. Both tables are regenerated from the archived XMLs on demand (`build-catalog`).

3.3 Glider categories

Soaring kinematics depend strongly on the aircraft, so the glider must be identifiable for any stratified analysis. Three broad categories exist, in increasing order of performance: **paragliders** (flexible canopy; slowest, glide ratio $\sim$8–11, tight turns), **hang gliders** (rigid-framed delta wing; faster, glide ratio $\sim$10–15), and **sailplanes** (rigid aircraft; fastest, glide ratio $\gtrsim$25). They differ several-fold in cruise speed and turn radius – which is why the reference study [4] reports its transport statistics separately for each.

In practice the category is set by the *data source*, since the public databases are organised by discipline. The FFVL runs the Coupe Fédérale de Distance as two parallel competitions – one for *paragliders* and one for *hang gliders* – so the present dataset spans *both* glider types: the glider-type stratum has two levels, and the cross-type comparison of the reference study [4] is therefore already possible between paragliders and hang gliders. The third category, *sailplanes*, is collected by different portals and is not yet in hand (Chapter 2); until it is added the cross-type analysis is limited to the two FFVL disciplines.

Within each discipline the flights can be further stratified by *performance class*, recorded in `aile_class`; the two disciplines use different class systems. FFVL labels *paragliders* on the EN/AFNOR certification ladder – EN A (“A ou 1”, most stable, recreational), EN B (“B ou

1-2”), EN C (“C ou 2”), EN D (“D ou 2-3”), the “CIVL Competition Class” (CCC) racing wings, plus tandem (“biplace”) and non-certified (“non homologuée”) entries. *Hang gliders* follow the FAI/CIVL class system, which in this data reduces to flexible wings (“Delta Class 1” and the intermediate “Delta Class Sport”) and rigid wings (“Rigide Class 2” and “Rigide Class 5”). In both cases the ladder tracks performance – higher classes fly faster and flatter – so it is the natural within-source stratification variable, playing the role that glider type plays across sources.

Two caveats bear on any use of `aile_class`. First, the field must be normalised before it can serve as a stratum: label variants are merged, and a small fraction of rows carry a blank or a placeholder (“0”) class, for which the class is recovered from the wing model (`aile`) where possible. Second, the class certifies the wing, not the pilot. Both points are handled at the pre-processing stage (Chapter 4).

3.4 Coverage and statistics

The statistics below describe the FFVL CFD dataset currently in hand; further sources may be added later (Chapter 2). It comprises 212,602 flights in total, of which 192,971 carry a GPS tracklog and 192,768 have been downloaded and verified, split across two disciplines.

The *paraglider* collection spans 27 seasons, from 1999–2000 to 2025–2026: 203,343 flights, of which 186,225 (91.6 %) carry a tracklog and 186,052 (99.9 %) are downloaded and verified (Table 3.1). The *hang-glider* collection spans 25 seasons, from 2001–2002 to 2025–2026: 9,259 flights, of which 6,746 (72.9 %) carry a tracklog and 6,716 (99.6 %) are downloaded and verified (Table 3.2). The hang-glider archive is roughly twenty times smaller, reflecting the smaller active population of the discipline.

In both disciplines the volume of flights grows by orders of magnitude over the period, reflecting the adoption of GPS loggers and online scoring: the early seasons contribute a handful of tracks each, while recent seasons contribute thousands.

Table 3.1: FFVL CFD *paraglider* flights per season (`parapente.ffvl.fr`): total declared, with a GPS track, and downloaded. Source: `data/paragliders/seasons_index.csv`.

Season	Flights	With GPS	Downloaded	Coverage (%)
1999–2000	10	10	10	100.0
2000–2001	832	1	1	100.0
2001–2002	1,150	3	3	100.0
2002–2003	1,996	123	123	100.0
2003–2004	1,470	337	337	100.0
2004–2005	1,765	574	574	100.0
2005–2006	2,162	894	894	100.0
2006–2007	2,386	1,016	1,016	100.0
2007–2008	3,081	1,667	1,667	100.0
2008–2009	3,974	2,614	2,613	100.0
2009–2010	4,171	3,059	3,059	100.0
2010–2011	4,872	4,185	4,169	99.6
2011–2012	6,744	6,075	6,065	99.8
2012–2013	7,873	7,106	7,075	99.6
2013–2014	9,804	9,157	9,127	99.7
2014–2015	13,793	13,293	13,254	99.7
2015–2016	13,982	13,703	13,696	99.9
2016–2017	15,760	15,534	15,528	100.0

Season	Flights	With GPS	Downloaded	Coverage (%)
2017–2018	14,917	14,682	14,676	100.0
2018–2019	16,702	16,517	16,508	99.9
2019–2020	10,966	10,868	10,863	100.0
2020–2021	9,500	9,483	9,480	100.0
2021–2022	14,552	14,507	14,502	100.0
2022–2023	11,590	11,561	11,560	100.0
2023–2024	10,372	10,362	10,360	100.0
2024–2025	10,906	10,885	10,884	100.0
2025–2026	8,013	8,009	8,008	100.0
Total	203,343	186,225	186,052	99.9

Table 3.2: FFVL CFD *hang-glider* flights per season (`delta.ffvl.fr`): total declared, with a GPS track, and downloaded. Source: `data/hang_gliders/seasons_index.csv`.

Season	Flights	With GPS	Downloaded	Coverage (%)
2001–2002	248	0	0	0.0
2002–2003	281	6	6	100.0
2003–2004	256	28	25	89.3
2004–2005	244	35	25	71.4
2005–2006	365	80	76	95.0
2006–2007	337	100	96	96.0
2007–2008	388	136	132	97.1
2008–2009	344	155	155	100.0
2009–2010	324	136	136	100.0
2010–2011	425	147	147	100.0
2011–2012	117	103	103	100.0
2012–2013	428	406	406	100.0
2013–2014	423	409	409	100.0
2014–2015	628	595	591	99.3
2015–2016	417	413	413	100.0
2016–2017	468	462	462	100.0
2017–2018	342	331	331	100.0
2018–2019	604	594	594	100.0
2019–2020	378	373	373	100.0
2020–2021	328	328	328	100.0
2021–2022	570	568	568	100.0
2022–2023	376	374	374	100.0
2023–2024	437	437	436	99.8
2024–2025	353	352	352	100.0
2025–2026	178	178	178	100.0
Total	9,259	6,746	6,716	99.6

3.5 Data quality and caveats

A few limitations are already known and will shape the analysis:

- **Flights without a track.** About 8 % of paraglider entries and about 27 % of hang-glider entries have no IGC file (older or manually declared flights); they carry metadata only.
- **A small residue of unrecoverable files.** A few flights advertise a track that the server does not actually serve as a valid IGC (broken or placeholder links); these are recorded as failures and excluded.
- **Heterogeneous logging.** Sampling rate, altitude source, and accuracy depend on the device; trajectories must be resampled and quality-filtered before any ensemble statistic is computed.
- **Altitude channel.** The barometric channel is adopted for the vertical dynamics (Chapter 4), but a sizeable minority of loggers record no pressure altitude – over a quarter of paraglider and roughly a sixth of hang-glider flights – which fall back to the GNSS altitude, flagged as such.
- **Glider metadata.** Flights are paragliders or hang gliders, tagged by source (Section 3.3); the class field `aile_class` uses a different system in each discipline and must be normalised before it is used to stratify.
- **Incomplete dates.** Some historical entries have placeholder dates (e.g. `0000-00-00`); these are kept but flagged.

These caveats motivate the pre-processing described in Chapter 4.

Chapter 4

Next steps

With the dataset in place and verified, the work proceeds as follows.

4.1 Trajectory pre-processing

Each IGC file is parsed into a sequence of fixes $(t_i, \varphi_i, \lambda_i, h_i)$ – UTC time, geodetic latitude and longitude, and altitude – by the `soaring.analysis.igc` parser, which reads the `B` records at their fixed character positions and returns both altitude channels. The pipeline then proceeds in a fixed order: (i) choose the altitude channel; (ii) clean individual fixes; (iii) trim the ground phases; (iv) filter out unfit whole flights; (v) convert to a local Cartesian frame; (vi) enforce a uniform sampling interval within each flight; and (vii) smooth and differentiate the trajectory. The output is a tidy, analysis-ready representation in a growing `soaring.analysis` sub-package.

The ordering follows one rule: work in the frame that makes each step exact and cheap. Steps (i)–(iv) act on the *raw geographic* coordinates, since the only geometric quantities they need – inter-fix speed and spatial extent – follow from the great-circle (haversine) distance between fixes, with no coordinate conversion. The conversion (v) to the metric East–North–Up frame (Sec. 4.1.5) is then done on the cleaned, trimmed track, with the origin at the take-off fix ($\mathbf{r}(0) = \mathbf{0}$), so the tangent-plane frame is built from good data and referred to the right origin. The final steps (vi)–(vii), which produce filtered position, velocity and acceleration, are naturally metric and so run *after* the conversion.

4.1.1 Choice of altitude channel

The vertical dynamics – vertical velocity, climb/sink, and the phase segmentation that rests on them – are taken from the *barometric* altitude rather than the GNSS altitude. The reason is one of signal quality in the frequency band that matters. The barometric (pressure) altitude has sub-metre relative precision and very little high-frequency noise; its errors are slow – a constant offset from the ICAO reference and a gradual drift with the synoptic pressure over a multi-hour flight – and slow errors are annihilated by the differencing that every dynamical quantity involves (a vertical velocity, a displacement increment, an autocorrelation). The GNSS altitude, by contrast, is geometrically referenced but vertically noisier: the vertical dilution of precision makes it 1.5–2 times worse than the horizontal, and multipath adds high-frequency jitter of a few metres – exactly the band that differencing *amplifies*. This is also why the variometers used in free flight derive the climb rate from the pressure sensor, not from GNSS.

Empirical confirmation. This is confirmed empirically on our data. Figure 4.1 compares the two channels. Panels (a)–(c) use a seeded random subsample of 400 flights per discipline – a deliberate choice, since the Welch PSD is an ensemble-average spectral *shape*, not a headline number, so a moderate, explicitly sized subsample is the appropriate and standard tool (Appendix B). Their power spectra agree at low frequency, where the real climbs and glides live, but towards the Nyquist frequency the GNSS spectrum flattens into a noise floor two to three orders of magnitude above the barometric one; for a single flight, the two raw channels (panel a) and their difference (panel b) show the same jitter directly. It is worth being explicit that this choice *departs from the reference study* [4], whose pipeline derives the vertical coordinate and vertical velocity from the GNSS altitude (it reads the barometric channel but does not use it); the divergence is deliberate and is one of the points re-examined here.

Several practical points follow.

One channel per flight. A trajectory uses a single altitude channel end-to-end; the barometric and GNSS values are never spliced within a flight, since a mid-flight switch between two channels offset by tens of metres would inject a spurious jump.

Source recorded. Which channel a flight uses is stored as a per-flight field – an `alt_source` column (`baro` or `gnss`) in the processed dataset, alongside duration, fix count and the like; the raw IGC files are never modified.

Availability. A flight is used with its barometric channel when it has one, and the few individual fixes with a missing barometric value are dropped rather than back-filled from GNSS. A logger with *no* pressure sensor writes the whole barometric channel as zero (the presence is essentially all-or-nothing, not a scattering of gaps); such a flight falls back to the GNSS altitude, unfiltered – a non-negligible minority of flights (Figure 4.1d).

Measuring the fallback rate. Because it is a headline number, its precision is exact rather than estimated: both archives are now full-censused – 17.5 % of 6,716 hang-glider flights and 30.1 % of 186,025 paraglider flights carry no barometric channel. This exactness is a byproduct, not extra work: the same full parse of every track is needed anyway for the flight-level filtering diagnostics of Sec. 4.1.4 (Figure 4.3), so the barometric-presence fraction is read off that one scan rather than measured separately. (A sized random sample – the standard proportion formula $n = z^2 p(1-p)/e^2$ – remains the right tool when a full census is not already being paid for elsewhere; it is not needed here.)

Is a mixed dataset a problem? Not much, for two reasons. The *horizontal* transport analysis – the primary result (MSD and its scaling) – does not use the altitude at all, so every flight is usable there regardless of channel. Altitude enters only the vertical velocity, hence the segmentation and the phase-resolved observables; there the two source groups are kept distinct through `alt_source` and their statistics checked for compatibility (Sec. 4.2). If the noisier GNSS-only flights were to bias a vertical statistic, that analysis is restricted to the barometric flights – still the large majority – while the horizontal analyses keep the full sample. Forcing GNSS on *all* flights, to avoid the mix, would be the wrong trade: it would needlessly degrade the vertical signal of the barometric majority to match the worst loggers.

4.1.2 Fix-level cleaning

Geometry on the raw coordinates. Cleaning, trimming and flight-level filtering all run *before* the Cartesian conversion (Sec. 4.1.5), directly on the geodetic angles (φ_i, λ_i). Distances between fixes therefore cannot be Euclidean; they are great-circle (haversine) distances on a

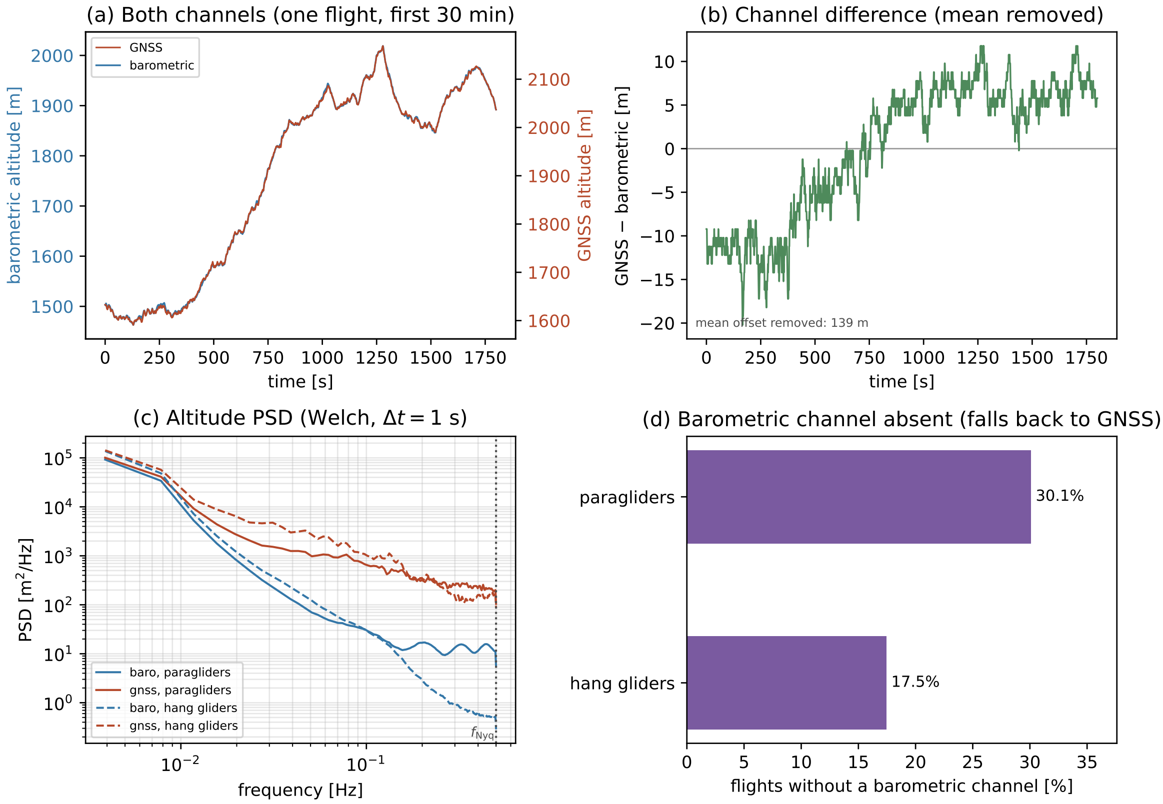


Figure 4.1: Barometric vs GNSS altitude noise. For one representative flight (the one with the most fixes in the subsample), over the same 30 min window: (a) the two raw channels, each on its *own*, independently auto-scaled y-axis (barometric left, GNSS right): the two curves then track almost on top of each other, since each is stretched to fill the panel with its own ∼500 m climb – the point of this panel is that shared shape, not their offset, which two independent axes no longer render as a literal vertical gap (that number is panel (b)'s job). Nor is this panel where the claim that GNSS is the noisier channel rests – at this timescale a climb of hundreds of metres dwarfs a jitter of a few metres on the plotted scale regardless of axis choice, so a single flight's raw trace is the wrong place to look for it; that evidence is panel (c). (b) Their difference GNSS−barometric (mean removed, *not* detrended): the offset and its drift, of order several tens of metres over the window, put on a single shared scale. That drift is a real, physically meaningful feature – plausibly the barometric channel's progressive departure from the ICAO standard atmosphere as altitude changes – not noise to be filtered out, so only the constant part of the offset is subtracted; its value for this flight (139 m) is annotated on the panel. Over the ensemble: (c) mean Welch power spectra of the two channels, per discipline (Appendix B; 212/400 paraglider and 114/400 hang-glider sampled flights actually contribute, after the common-cadence selection described there) – here, and only here, does the two-to-three-orders-of-magnitude gap between the channels at high frequency make the noise-floor claim; the dotted line is the Nyquist frequency $f_{\text{Nyq}} = 1/(2\Delta t)$, the highest frequency the sampling can resolve. It is a single line because the pooled flights are all resampled to the same modal Δt before averaging, so they share one Δt – hence one f_{Nyq}. (d) Fraction of flights whose barometric channel is absent, per discipline; see the main text for how these were measured and their precision.

sphere of mean radius $R_\oplus = 6{,}371\,\mathrm{km}$. For two consecutive fixes,

$$\Delta s_i = 2R_\oplus \arcsin \sqrt{\sin^2\frac{\varphi_{i+1}-\varphi_i}{2} + \cos\varphi_i \cos\varphi_{i+1}\, \sin^2\frac{\lambda_{i+1}-\lambda_i}{2}}\,, \tag{4.1}$$

from which the horizontal speed between the fixes is $v_{xy,i} = \Delta s_i/(t_{i+1}-t_i)$ and, from the barometric altitude, the vertical speed is $v_{z,i} = (h_{i+1}-h_i)/(t_{i+1}-t_i)$. Two whole-flight quantities used below follow from the same distance: the flown *path length* $L = \sum_i \Delta s_i$ and the *extent* $D = \max_i \Delta s(\mathrm{fix}_0, \mathrm{fix}_i)$, the farthest a fix gets from take-off. The spherical approximation is ample here – the ellipsoidal correction to an inter-fix distance is $\lesssim 0.3\,\%$, far below the GPS noise – and it avoids converting coordinates before the trajectory has been cleaned.

Two layers of cleaning. Individual fixes that are physically impossible or corrupted must be removed without discarding the whole flight. This happens in two layers, at two different stages of maturity. The first, *format validity*, is already implemented: the `soaring.analysis.igc` parser rejects, at decode time, any fix that does not decode to a *legal* record – an invalid UTC time, or a latitude/longitude whose arc-minutes, hemisphere letter or resulting value are not a valid encoding (Chapter 3) – since such a fix carries no meaningful position or time at all; this is covered by unit tests. The second, *dynamics plausibility*, is design work still to be implemented: Table 4.1 lists the criteria – fixes that decode to a legal record but are implausible given the trajectory, such as an impossible speed jump (via v_{xy} from (4.1)), an out-of-range altitude, or a stale duplicate. The bounds are not guessed but read off the *per-fix* distributions of these quantities on the real tracks (Figure 4.2): each sits in the sparse, physically-implausible tail – a GPS error, not signal – and removes only a negligible fraction of fixes, so the cut cleans the trajectory without eroding it. This mirrors the flight-level audit of Sec. 4.1.4, with one twist: a fix-level bound drops single fixes from an otherwise good flight, so what matters is the fraction of *fixes* it removes, not the number of flights it touches – a bound may legitimately touch many flights. The numeric values live in the config (`configs/preprocessing.yaml`, `FixLevelThresholds`); the routine that walks a trajectory applying them, fix by fix, does not yet exist – building it is part of this stage. The checks apply in the order listed, each acting only on the fixes the previous ones spared.

Two points a table cannot convey. First, the IGC validity flag (`A`/`V`) is not a separate criterion but is *subsumed* by the missing-altitude rule applied to the adopted channel. The flag certifies the GNSS altitude only: a `V` fix has the GNSS altitude zeroed but keeps a valid position and, if present, a valid barometric altitude (Chapter 3). Its effect is therefore channel-dependent – on a barometric flight a `V` fix is kept (the barometric value is intact), whereas on a GNSS-fallback flight it has no usable altitude and is dropped as a missing-altitude fix – and needs no handling of its own. Second, inter-fix time gaps are deliberately *not* a fix-level criterion. A gap is not itself an error – short ones interpolate cleanly, long ones do not – and, unlike a speed or altitude spike, it is a property of the sampling rather than of a single fix. It is therefore handled once, at the flight level (Sec. 4.1.6): a flight whose largest gap is too large *relative to its own native cadence* is excluded there. A second, absolute gap bound at the fix level would only duplicate that test with a differently-scaled threshold.

4.1.3 Trimming of ground phases

The logger starts recording before take-off and stops after landing; these ground phases must be stripped before any kinematic analysis. Detection uses the *horizontal* speed v_{xy} alone (from (4.1)) – the vertical speed plays no role here: the airborne segment begins when v_{xy} rises above $5\,\mathrm{m\,s^{-1}}$ and stays above it for at least $30\,\mathrm{s}$, and ends symmetrically when it falls

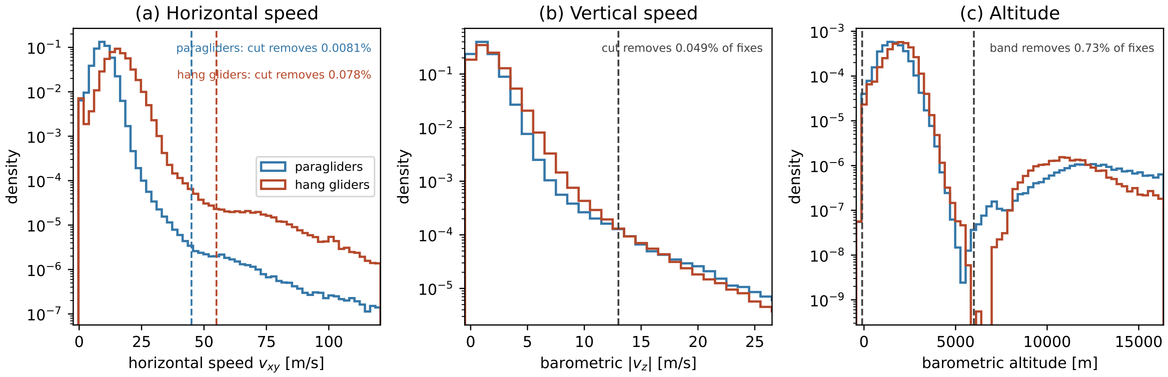


Figure 4.2: Fix-level cleaning, audited on the data. Per-*fix* distributions of the quantities the dynamics-plausibility bounds test – (a) horizontal speed v_{xy}, (b) barometric $|v_z|$, (c) barometric altitude – pooled over a seeded random sample of 15,000 paraglider (126,723,499 consecutive-fix pairs, $\sim$73 s to compute) and all 6,716 hang-glider flights (a full census at this population size, 37,915,334 pairs, $\sim$25 s) – deliberately large, so that the sparse tail where the bounds act is resolved well enough to tell real dynamics from a rare logger artifact, at a computational cost of under two minutes. Dashed lines mark the adopted bounds (Table 4.1). In (a), horizontal speed, the cut is one per discipline and colour-matched to it, since paragliders and hang gliders have different horizontal-speed envelopes, with the fraction of fixes each would remove annotated per discipline; in (b)/(c) the cut/band is shared, since neither vertical speed nor altitude is meant to track a discipline-specific limit. The y-axis is logarithmic so that tail is visible; in (a)/(b) the x-range is deliberately wide (out to each discipline's own 99.99th percentile) rather than cropped just past the bound, so the excluded tail is actually shown, not merely implied by the removed-fraction number – panel (c) keeps a narrower range, since its cuts sit far out in an otherwise tight, unimodal distribution and widening it the same way would mostly shrink the informative part of the panel. Panel (a) is why the speed bounds are per-discipline and not a shared value: at this sample size, hang gliders show a distinct, much more slowly-decaying plateau from about 55 m s^{-1} to 90 m s^{-1} sitting on top of the steeply-decaying real distribution – a sustained *ground* speed of 200 km h^{-1} to 320 km h^{-1} has no credible airborne explanation for a hang glider, so this reads as a real, distinct artefact population (most likely a logger/GPS pathology), not rare-but-genuine manoeuvres. Placing the bound is a choice between two candidates: the point where the decay first visibly *slows* ($\approx$40 m s^{-1}) and the point where it actually *stops*, i.e. the bin-to-bin ratio of a fine histogram settles at $\approx$1 ($\approx$55 m s^{-1}). The former still includes a shallow-but-real stretch of decreasing density – rare, wind-assisted glides or steep spirals do not vanish abruptly – so only the latter is a safe marker of "this is now the artefact, not the tail of real dynamics"; 55 m s^{-1} is adopted for that reason. Paragliders show the analogous pattern at a lower speed, adopted at 45 m s^{-1}. Panel (b) would be a comb, not a smooth curve, on a naively-binned axis, because barometric altitude is logged to the metre: differencing an integer-metre series over a 1 s step can only return an integer number of m s^{-1}, and an arbitrarily placed fine grid of bins then picks up an inconsistent share of each integer's spike. One bin per integer (used here) removes that artefact instead of just re-sampling it. The comb would not in any case be perfectly clean: not every flight samples at 1 s (Figure 4.7 shows hang gliders spread across several native rates), and differencing over a longer native step returns a finer-grained set of values (e.g. multiples of 0.2 m s^{-1} at a 5 s cadence) that partially fill the gaps once every cadence is pooled together. Checked directly: among fixes from 1 s-native flights, about 99% land within 0.05 m s^{-1} of an integer; among fixes from every other native cadence, only 34–42% do.

Quantity	Reject / flag the fix when	Rationale
Missing altitude	the adopted-channel value is absent	no usable vertical coordinate (Sec. 4.1.1)
Duplicate time	two fixes share the UTC second	keep the first (logger double-write)
Altitude range	outside $[-100, 6000]$ m	sensor fault, allowing for the channel's own offset
Horizontal speed (paragliders)	$v_{xy} > 45\,\mathrm{m\,s^{-1}}$ between consecutive fixes	GPS position jump
Horizontal speed (hang gliders)	$v_{xy} > 55\,\mathrm{m\,s^{-1}}$ between consecutive fixes	GPS position jump
Vertical speed	$\lvert v_z \rvert > 13\,\mathrm{m\,s^{-1}}$	beyond the soaring climb/sink range

Table 4.1: Fix-level cleaning criteria, applied in the order listed. Each bound marks the physically-implausible tail of its per-fix distribution (Figure 4.2), so it removes GPS errors rather than signal; the values are set in `configs/preprocessing.yaml` (never hard-coded). Horizontal speed is set one per discipline – paragliders and hang gliders have different horizontal-speed envelopes (the present dataset, Chapter 3), so a single shared bound is either too loose for the slower type or clips real dynamics of the faster one; a `"sailplanes"` entry will be added once that source is in. Vertical speed and the altitude range are *not* split by discipline: the two disciplines' vertical-speed distributions are close enough that splitting it buys nothing (Figure 4.2), and the altitude bound is about the barometric sensor's plausible reading range, not glider performance; both apply to the adopted channel (Sec. 4.1.1). The lower altitude bound is -100 m rather than 0 because the barometric channel is not QNH-corrected: its offset from true altitude (Figure 4.1, of order 100 m in the flight shown there) means a take-off at sea level can legitimately read slightly negative, so the bound must clear that offset and not just true altitude. Inter-fix gaps are handled once, at the flight level (Sec. 4.1.6), not with a second absolute bound here.

back below. The sustained-speed requirement prevents a gust or a GPS glitch on the ground from being mistaken for take-off; both the threshold and the persistence affect what fraction of each trajectory is retained, so they are made explicit – and, like every threshold, set in `configs/preprocessing.yaml`, not hard-coded.

4.1.4 Flight-level filtering

After cleaning and trimming, some flights are still unfit for the ensemble analysis and are dropped whole. The step is delicate: too aggressive a cut biases the transport statistics – a high minimum duration, for instance, preferentially keeps successful long flights and inflates the apparent persistence of the MSD. The flight-level filtering therefore uses a **minimal-threshold strategy**: for each criterion, the threshold is set to the smallest value that still clears non-genuine flights (sled runs, tows, aborted or truncated logs), so the ensemble is disturbed as little as possible. Once chosen, each cut is verified by the full-distribution diagnostics in Fig. 4.3: it sits in the empty tail of its distribution, where moving it changes almost nothing. All flight-level quantities are computed directly from the parsed tracks; the thresholds follow the minimal-threshold strategy just described. Two track quantities gate a flight, plus a metadata stratifier:

(1) *Recorded duration* $T \geq 40$ min. A cross-country flight spans many climb–glide cycles; shorter logs are sled runs or aborted attempts. The cut is deliberately *decoupled* from the maximum MSD lag – the ensemble at lag t already uses only flights with $T > t$, and the aging in T is diagnosed by stratifying in T – so large-lag analyses restrict to the long-duration stratum rather than raising this global cut.

(2) *Flown path length* $L \geq 30$ km, with $L = \sum_i \Delta s_i$ from the great-circle steps of (4.1). This

removes tows and local hops that never became a cross-country flight. It is a minimal cut: a real XC flight covers many tens to hundreds of kilometres, far above 30 km.

(3) *Glider class* (metadata). For class-stratified analyses, flights whose `aile_class` cannot be normalised onto the discipline's system are dropped from the class-resolved sub-analyses but kept in the pooled one. Stratification by glider *type* (paraglider vs hang glider) is already possible; the sailplane stratum follows once that source is in hand.

A separate minimum-fix-count cut would be redundant: at $T \geq 40$ min even the slowest logger (5 s) already records several hundred fixes, well above what the kinematics need. A further criterion – sampling regularity – is applied later, where a flight's time base is examined (Sec. 4.1.6). All thresholds live in `configs/preprocessing.yaml` (never hard-coded).

Figure 4.3 shows, for every discipline and computed on the *full* dataset (every track, no sub-sampling), the distributions of the two track cuts and, beside each, the fraction of flights that cut *alone* would retain – its marginal effect, not a cascade. Both cuts sit far out in the tail, so the retained fraction is insensitive to their exact value; the diagnostics must be re-examined for the less curated further sources.

4.1.5 From geographic to local Cartesian coordinates

As the final pre-processing step, the cleaned, trimmed and resampled track is mapped from geographic to local Cartesian coordinates. Latitudes and longitudes are angles on a curved surface and cannot be differenced directly to obtain distances; the mapping proceeds in two steps, following the convention of Vilpellet et al. First, each fix is converted to Earth-Centred, Earth-Fixed (ECEF) coordinates on the WGS84 ellipsoid (semi-major axis $a = 6{,}378{,}137$ m, flattening $f = 1/298.257223563$, $e^2 = f(2-f)$) [6],

$$N(\varphi) = \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi}}, \qquad \begin{aligned} X &= (N + h)\cos\varphi\cos\lambda, \\ Y &= (N + h)\cos\varphi\sin\lambda, \\ Z &= \big(N(1 - e^2) + h\big)\sin\varphi, \end{aligned} \tag{4.2}$$

where $N(\varphi)$ is the prime-vertical radius of curvature and h the ellipsoidal height, measured along the ellipsoid normal.

A point of care: the latitude φ in (4.2) is the *geodetic* latitude – the angle between the equatorial plane and the ellipsoid *normal* at the point (Figure 4.4), *not* the geocentric latitude ψ, which is the angle to the line from the Earth's centre O. On an ellipsoid the two differ, because the normal does not pass through O (Figure 4.4): the gap $\varphi - \psi$ reaches $\approx 0.19°$ (about $11'$) near $45°$, which if the two were confused would misplace a fix by up to $\sim$20 km. We use the geodetic latitude throughout, and no conversion is needed: WGS84 – hence the GPS receiver and the IGC `B` records (Chapter 3) – reports geodetic latitude, and (4.2) with $N(\varphi)$ is precisely the geodetic-to-ECEF transform that consumes it. The longitude λ carries no such ambiguity: it is the same angle in both conventions, since the ellipsoid is a surface of revolution about the polar axis. The geodetic latitude and longitude are exactly the (φ, λ) that orient the local frame below.

Second, taking the first fix $(\varphi_0, \lambda_0, h_0)$ of the flight – the take-off fix, since the ground phase has already been trimmed – as the origin, the ECEF displacement $\mathbf{\Delta X} = \mathbf{X} - \mathbf{X}_0$ is rotated into the local East–North–Up (ENU) frame tangent to the ellipsoid at that point,

$$\begin{pmatrix} E \\ N \\ U \end{pmatrix} = \begin{pmatrix} -\sin\lambda_0 & \cos\lambda_0 & 0 \\ -\sin\varphi_0\cos\lambda_0 & -\sin\varphi_0\sin\lambda_0 & \cos\varphi_0 \\ \cos\varphi_0\cos\lambda_0 & \cos\varphi_0\sin\lambda_0 & \sin\varphi_0 \end{pmatrix} \mathbf{\Delta X}. \tag{4.3}$$

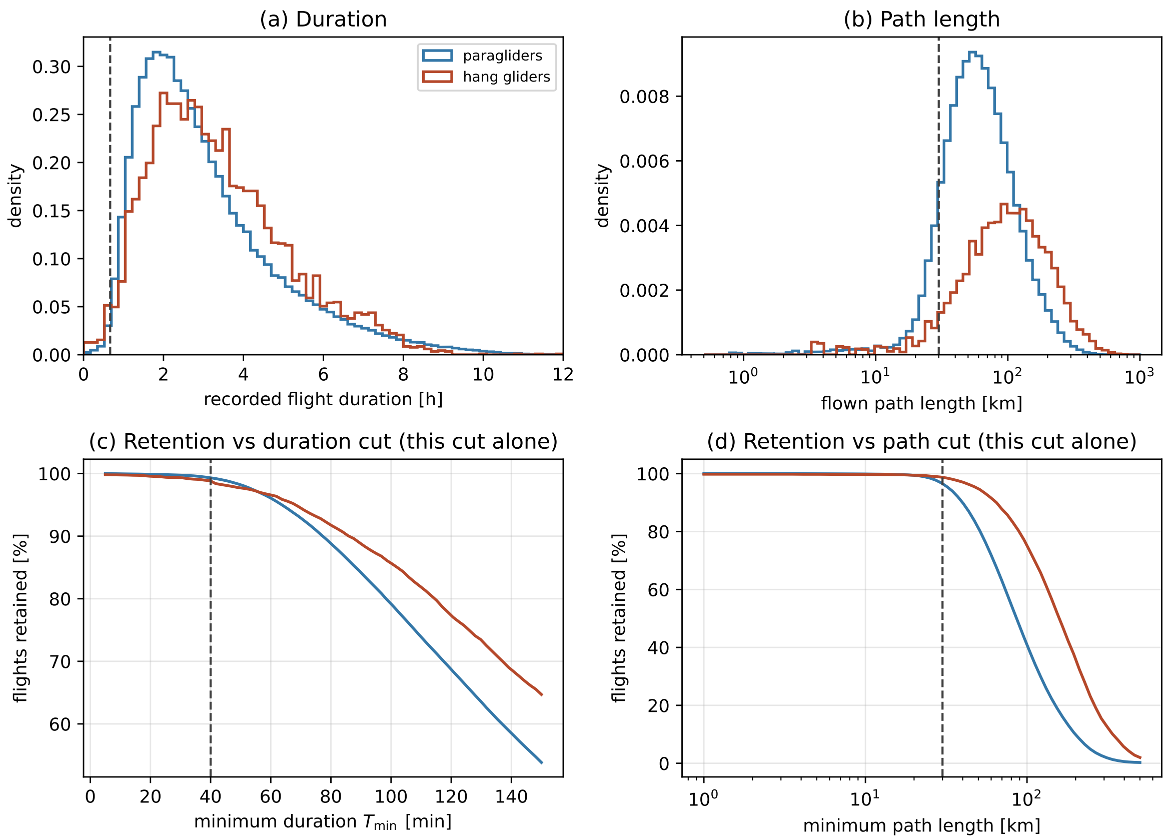


Figure 4.3: Flight-level filtering diagnostics, computed on the *full* dataset – every parsed track, per discipline (paragliders and hang gliders; sailplanes later) – not from the declared metadata and not sub-sampled. Top: distributions of (a) recorded flight duration and (b) flown path length. Bottom: the fraction of flights retained versus (c) the duration cut and (d) the path-length cut, each computed for *that cut alone* (marginal, not cascaded). Dashed lines mark the adopted cuts ($T \geq 40\,\text{min}$, $L \geq 30\,\text{km}$), which sit far out in the tail. On the full census (186,025 paraglider and 6,716 hang-glider tracks) the median duration and path length are 2.7 h/86 km for paragliders and 3.1 h/158 km for hang gliders – the latter flying faster and farther.

The ENU frame *is* the local Cartesian frame: its horizontal plane (E, N) is the "north–east" plane and U is the local vertical, so there are not two distinct coordinate systems but one, reached through the intermediate (global) ECEF representation. Over the spatial extent of a single flight (tens of kilometres) the tangent-plane approximation is accurate to well below the GPS noise. What the rotation matrix actually *does* – turning a vector's global ECEF components into local East/North/Up ones – is shown in Figure 4.5.

The altitude h that enters this transform is the adopted barometric channel (Sec. 4.1.1); changing channel would move (E, N) by a relative $h/(N + h) \sim 10^{-5}$, i.e. negligibly, so the horizontal coordinates are insensitive to the altitude source. The vertical coordinate $z = U$ is therefore, up to the small deterministic tangent-plane curvature term, just the barometric height referred to the take-off value, $z \approx h - h_0$ – exactly the quantity whose dynamics were argued above to be cleanest.

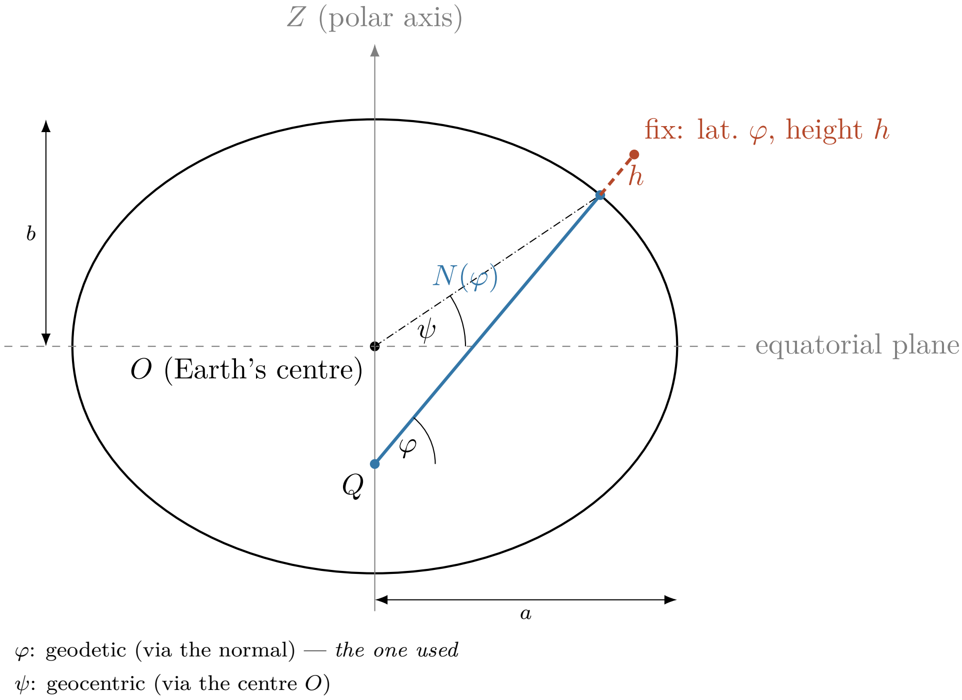


Figure 4.4: Geodetic latitude and height on the reference ellipsoid, in meridian cross-section (flattening exaggerated for visibility; true WGS84 flattening $f \approx 1/298$ would look indistinguishable from a circle at this scale): the semi-major and semi-minor axes (a, b), the geodetic latitude φ – the angle between the ellipsoid normal and the equatorial plane – and the height h above the ellipsoid along that normal. The normal does *not* pass through the centre O except at the pole and the equator; $N(\varphi)$ is precisely the distance, along the normal, from the ellipsoid surface to the point Q where it crosses the polar axis. Because of this, the geodetic latitude φ (via the normal, the one WGS84/GPS reports and the one used here) differs from the *geocentric* latitude ψ (dash-dotted, the angle of the radius O–P): $\psi < \varphi$ off the pole and equator. The third geodetic coordinate, longitude λ, is not visible in this cross-section (it is the angle *around* the polar axis, out of the page here) and is shown instead in Figure 4.5, together with how (φ, λ) orient the local frame used throughout.

4.1.6 Uniform sampling within a flight

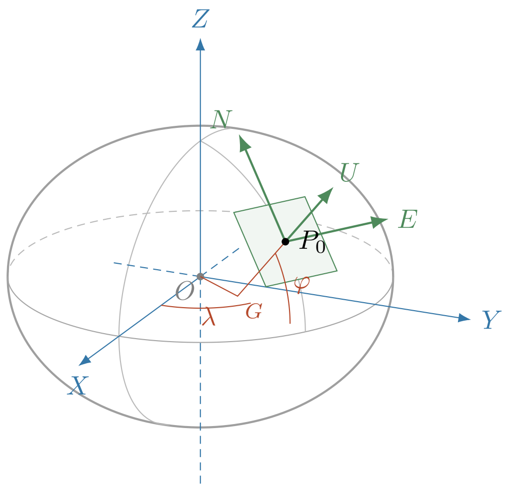
The smoothing and differentiation below assume a *uniform* sampling interval, so each surviving flight's time base is examined next. Loggers record at a fixed nominal rate – paragliders almost all at 1 s, hang gliders spread more widely across several rates up to 10 s (Fig. 4.7) – but the realised series can carry jitter and drop-outs. We keep each flight at its *native* rate rather than resampling to a common cadence: a common grid would either upsample slow loggers (inventing information) or throw away the resolution of fast ones, and it is unnecessary, since the derivative operator below is given the flight's own Δt (Sec. 4.1.7). What matters is uniformity *within* a flight, not across flights.

Per flight, let Δt be the median inter-fix interval. Three cases arise:

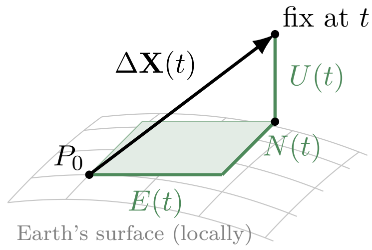
Uniform. All intervals equal Δt to tolerance: the series is used as is.

Mildly irregular. Isolated jitter or single drop-outs – the largest gap is at most a few Δt and the missing fraction is small: the flight is resampled onto the uniform grid at its native Δt, interpolating across the small gaps only, and the interpolated fraction is recorded.

Badly sampled. A gap longer than a set bound, or too large a missing fraction: the flight



(a)



(b)

Figure 4.5: What the rotation matrix does, reproduced in our notation. (a) The global ECEF frame (X, Y, Z) in blue, centred at Earth's centre O; the two red angles that locate the take-off fix P_0 – longitude λ, measured in the equatorial plane from the prime meridian (X), and the geodetic latitude φ, the tilt of the ellipsoid normal, measured at the point G where that normal meets the equatorial plane; and the local ENU frame (E, N, U) in green, tangent to the ellipsoid at P_0. The globe is drawn as an oblate ellipsoid (flattening exaggerated, as in Figure 4.4), which makes visible that U follows the geodetic normal G–P_0 and *not* the radius from O. The green arrows are the very directions the rotation matrix sends X, Y, Z onto – its three rows, as unit vectors. (b) The same tangent plane in close-up (N receding into the page). A later fix's ECEF displacement $\Delta\mathbf{X}(t) = \mathbf{X}(t) - \mathbf{X}_0$ is the space diagonal (black) of the box whose edges are its ENU components $E(t)$, $N(t)$, $U(t)$ – exactly the three rows of the matrix above, applied to $\Delta\mathbf{X}$.

is *excluded* (and logged). Interpolating across a long hole would fabricate dynamics the smoothing could not tell from real motion; splitting such a flight at its gaps into independently analysed segments is a possible future refinement.

The gap bound and the maximum missing fraction are the two thresholds of this step (in `configs/preprocessing.yaml`); the exclusion is the further flight-level cut promised in Sec. 4.1.4, and Figure 4.6 makes its retained fraction auditable in the same way as the duration and path-length cuts. One caveat carries downstream: a few observables are intrinsically Δt-dependent (a finite-difference velocity, a turning angle), so where they are compared across flights the ensemble is stratified by rate or restricted to the dominant one, rather than pretending a single cadence.

4.1.7 Smoothing and differentiation

Velocity and acceleration drive the kinematics and the segmentation, but differencing raw GPS positions amplifies the high-frequency noise characterised in Sec. 4.1.1. A **Savitzky–Golay** filter solves both problems at once: it fits a low-order polynomial to a sliding window by least squares and reads off that polynomial – or its derivatives – at the window centre, returning smoothed position, velocity and acceleration ($\text{deriv} = 0, 1, 2$) in one pass per component. The derivative outputs are scaled by the flight's own Δt (Sec. 4.1.6), so the multi-rate fleet needs no common grid, and the window never crosses a flight boundary.

The filter has two hyperparameters – the window length w (odd, in samples) and the polynomial order p – fixed by a *quantitative*, noise-matched procedure, not by eye:

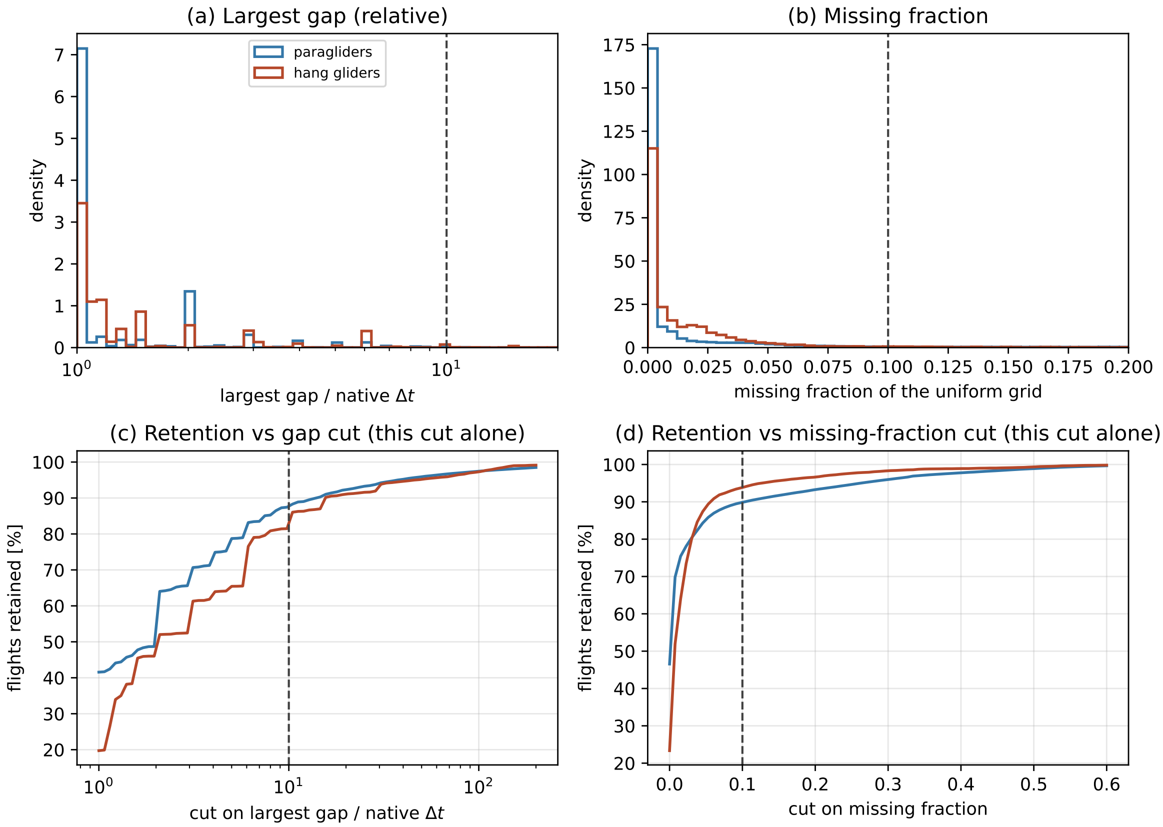


Figure 4.6: Sampling-regularity diagnostics, computed on the full dataset per discipline, on the *same* track scan as Figure 4.3 (so every parsed flight is included, not only the ones that survive the duration/path-length cuts). (a)/(b) Distribution of the largest single gap (relative to that flight's own native Δt) and of the missing fraction of a uniform grid at that interval. (c)/(d) The fraction of flights each cut would retain *on its own* (marginal, not cascaded), with the adopted thresholds marked.

(i) From the ensemble Welch power spectrum of the raw ENU components on a sample of flights (the diagnostic of Sec. 4.1.1 applied to E, N, U), read the frequency f_c of the knee where the dynamical signal meets the flat GPS noise floor. Its reciprocal is a smoothing timescale $\tau_c = 1/f_c$: dynamics slower than τ_c are signal, faster fluctuations are noise.

(ii) Fix p on a physical argument: to represent local curvature and give a clean second derivative the fitted polynomial must be at least quadratic, while a low order keeps the pass-band flat; $p = 3$ works and the result is insensitive to p within $\{2, 3, 4\}$.

(iii) Set the window per flight from the *timescale*, $w = \mathrm{odd}(\tau_c/\Delta t)$, so the physical smoothing scale is the same across the fleet regardless of rate.

(iv) Treat the vertical separately from the horizontal: the vertical GPS noise is 1.5–3× worse (dilution of precision), so its knee f_c, and hence its window, differ from those of (E, N).

The choice is validated on a held-out sample: the residual (raw minus smoothed) spectrum should be flat above f_c – confirming that noise, not signal, was removed – the acceleration distribution should stay within physical bounds, and the results should be stable under a one-step change of w. Both hyperparameters are thus anchored to a measured property of the data rather than to a guess; the order p and the two smoothing timescales τ_c (horizontal

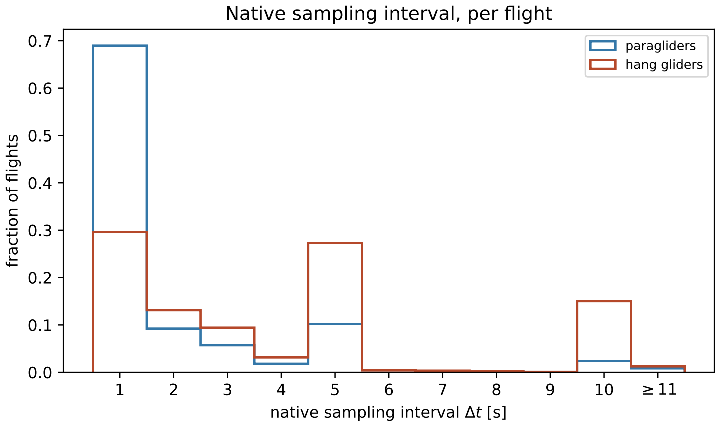


Figure 4.7: Native sampling interval Δt (median inter-fix time) per flight, on the full dataset per discipline. Δt is, in practice, a *discrete* quantity, not a continuous one: it lands on an exact whole second for 99.9% of paragliders and 100% of hang gliders (a logger reports at one of a handful of fixed configured rates), so bins are one second wide and centred on each integer – each bar is the fraction of flights at exactly that rate, not a density (with unit-width bins the two coincide numerically, but only "fraction" describes what is actually being counted). The fleet is multi-rate: paragliders are overwhelmingly (69%) at 1 s, whereas hang gliders spread across several rates – 1 s (30%), 5 s (27%), 10 s (15%), 2 s (13%), and a long thin tail beyond 11 s (last bin, aggregated). A single common cadence would waste the fast loggers; instead each flight keeps its own Δt, which the smoothing step consumes directly (Sec. 4.1.7).

and vertical) are stored in `configs/preprocessing.yaml`, to be fixed once the ENU power spectra are measured.

4.1.8 Notation

Throughout, $\mathbf{r}(t) = (x(t), y(t)) = (E(t), N(t))$ denotes the *horizontal* position in this local frame and $z(t) = U(t)$ the (barometric) altitude. By construction each trajectory starts at the horizontal origin, $\mathbf{r}(0) = \mathbf{0}$, with the clock reset to $t = 0$ at take-off, so t is the elapsed flight time. The vertical is *not* re-zeroed: the take-off height is retained, so the analysis keeps both the increment that drives the dynamics (invariant to any offset) and the absolute altitude for interpretation (potential energy, convective-boundary-layer structure). The horizontal velocity is $\mathbf{v}(t) = \dot{\mathbf{r}}(t)$, with magnitude the horizontal speed $v_{xy} = |\mathbf{v}|$; the vertical velocity is $v_z = \dot{z}$. The horizontal displacement over a lag t is $\Delta\mathbf{r}(t) = \mathbf{r}(t_0 + t) - \mathbf{r}(t_0)$.

4.2 Preliminary characterization

Before the main transport analysis it is useful to run a set of lightweight checks that validate basic assumptions, reveal potential stratification needs, and characterize the dataset at the level of individual trajectories. These do not require segmentation and are conceptually simpler than the transport observables, but they are prerequisites for a well-founded analysis.

Trajectory statistics. For each flight, computed from its parsed track: airborne duration T, total horizontal path length, number of valid fixes and native sampling interval Δt. Their ensemble distributions – the filtering-relevant ones in Fig. 4.3 – characterise what the dataset contains, flag outliers, and, through Δt, set the per-flight smoothing window (Sec. 4.1.7). In the present data durations centre on a few hours and track lengths on $\sim 10^4$ fixes, so the flight-level cuts fall in an essentially empty tail.

Isotropy check. The scaling of the 1D marginal propagator (Section 4.3) requires the process to be isotropic. Competition flights have preferred directions (race goal, orography, prevailing winds), so isotropy must be tested rather than assumed.

The primary test is the **per-component variance ratio**: for an isotropic process $\langle E(t)^2\rangle = \langle N(t)^2\rangle = \mathrm{MSD}(t)/2$ for all t, so the ratio $\langle E(t)^2\rangle/\langle N(t)^2\rangle$ should be identically 1. Plotting it as a function of t reveals both the degree and the scale-dependence of any anisotropy. A complementary check is a Kolmogorov–Smirnov test comparing the empirical distributions of $E(t)$ and $N(t)$ at a few selected lags.

If isotropy fails, the two marginals $P(E(t) = x)$ and $P(N(t) = x)$ are analysed *separately*: this is both the natural fallback and, in itself, informative – unequal scaling exponents $H_E \neq H_N$ or equal exponents but different scaling functions signal geometrically anisotropic transport.

Compatibility across strata. The dataset spans more than two decades, two glider types (paragliders and hang gliders), and a range of performance classes (Chapter 3) – from recreational EN A/B paraglider wings to CCC racing wings and the flexible/rigid hang-glider classes – whose kinematics differ substantially; sailplanes will later add a third type. Before treating all flights as draws from a single statistical process – i.e. before computing ensemble averages over the pooled dataset – the subpopulations must be checked for statistical compatibility.

A lightweight way to do this: for each stratum (e.g. a single wing class, or the flights of a single season) compute the per-component displacement variance $\langle E(t)^2\rangle$ as a function of the elapsed time t and overlay the curves on one plot. If they coincide, the transport statistics are compatible and pooling is justified. If one stratum grows systematically faster or with a different shape – competition wings, say, cover ground faster than recreational ones – the strata must be analysed separately, and that difference is itself a result about the universality of the transport regime.

Spatial coverage. A map of take-off coordinates and displacement end-points at fixed t reveals geographic clustering (France has distinct orographic zones: Alps, Pyrenees, Massif Central, plains). If flights from mountainous terrain differ markedly from flatland flights, geographic stratification may be needed alongside temporal and aeronautical stratification.

4.3 Observables on the whole trajectory

A first group of observables requires *no* segmentation: it can be measured directly on the full trajectories and already probes the transport at the global level. For each, we state what it is, how it is computed, and what it is for.

Transport and diffusion.

Mean-squared displacement (MSD). With every flight starting at the origin ($\mathbf{r}(0) = \mathbf{0}$), the *ensemble-averaged* MSD is the mean squared horizontal distance reached after an elapsed time t,

$$\mathrm{MSD}(t) = \langle\, |\mathbf{r}(t)|^2 \rangle, \tag{4.4}$$

where $\langle\cdot\rangle$ is an average *over flights* (an ensemble average) at fixed t. Its growth law $\mathrm{MSD}(t) \sim t^\alpha$ fixes the transport regime ($\alpha = 1$ Brownian, $\alpha > 1$ super-diffusive, $\alpha < 1$

sub-diffusive; equivalently the Hurst exponent $H = \alpha/2$). This is the primary test for anomalous transport.

Time-averaged MSD (TA-MSD). A second MSD estimator is built entirely *within* a single flight. For flight k of duration T_k, one slides a window of width t along the trajectory $\mathbf{r}^{(k)}$ and averages the squared displacement over all starting times t_0 *inside that flight*:

$$\overline{\delta^2}^{(k)}(t) \;\equiv\; \frac{1}{T_k - t}\int_0^{T_k - t} \left|\mathbf{r}^{(k)}(t_0 + t) - \mathbf{r}^{(k)}(t_0)\right|^2 \mathrm{d}t_0. \tag{4.5}$$

The average over t_0 here is a *time* average internal to flight k; it has nothing to do with the ensemble average $\langle \cdots \rangle$ over different flights used in the MSD. Because a single-flight curve $\overline{\delta^2}^{(k)}(t)$ is noisy, the operationally useful quantity is its mean over all flights of comparable duration T:

$$\mathrm{TAMSD}(t, T) \;\equiv\; \left\langle \overline{\delta^2}^{(k)}(t) \right\rangle_{T_k \approx T}. \tag{4.6}$$

Ergodic case. For an ergodic process, $\mathrm{TAMSD}(t, T) \to \mathrm{MSD}(t)$ as $T \to \infty$, independently of T, and the individual values $\overline{\delta^2}^{(k)}(t)$ concentrate around the common limit (the distribution collapses to a delta function).

Non-ergodic case: two independent signatures.

(i) *Aging.* $\mathrm{TAMSD}(t, T)$ differs from $\mathrm{MSD}(t)$ and retains a systematic dependence on T: flights of different duration yield different curves. This is a strong indicator of non-ergodicity, but it is sufficient, not necessary – some non-ergodic processes show only a weak T-dependence.

(ii) *Non-self-averaging.* The distribution of $\overline{\delta^2}^{(k)}(t)$ across trajectories remains broad even as $T \to \infty$: individual flights give systematically different TA-MSD values and the distribution does not collapse. This is the more fundamental signature: it persists even when aging is weak.

We therefore monitor both $\mathrm{TAMSD}(t, T)$ and the full distribution of $\overline{\delta^2}^{(k)}(t)$ across trajectories at fixed t and T. The canonical illustration – a CTRW with power-law waiting times – is worked out analytically in Appendix A, where both EA-MSD≠TA-MSD and the T-dependence are derived explicitly.

Propagator. Since each flight starts at the origin, $\mathbf{r}(0) = \mathbf{0}$, the position at time t and the displacement from the start coincide: $\mathbf{r}(t) = \mathbf{r}(t) - \mathbf{r}(0)$. We therefore define the propagator as the probability density of the position vector $\mathbf{x} \in \mathbb{R}^2$ at time t,

$$P(\mathbf{x}, t)\, \mathrm{d}^2x = \Pr\big(\mathbf{r}(t) \in [\mathbf{x},\, \mathbf{x} + \mathrm{d}\mathbf{x}]\big). \tag{4.7}$$

For a *self-similar* process with exponent H the propagator obeys the scaling form

$$P(\mathbf{x}, t) = t^{-2H}\, G\!\left(\frac{\mathbf{x}}{t^H}\right), \tag{4.8}$$

where $G : \mathbb{R}^2 \to \mathbb{R}_{>0}$ is a time-independent scaling function (normalised, $\int G\, \mathrm{d}^2u = 1$).

In practice one works with the **one-dimensional marginal** obtained by projecting onto a single Cartesian component x,

$$P(x, t) \equiv \int_{-\infty}^{+\infty} P((x, y), t)\, \mathrm{d}y. \tag{4.9}$$

Assuming the process is *isotropic*, $G(\mathbf{u}) = G(|\mathbf{u}|)$, the integral can be evaluated by the substitution $y = t^H v$:

$$P(x,t) = t^{-2H} \int_{-\infty}^{+\infty} G\left(\sqrt{\left(\tfrac{x}{t^H}\right)^2 + v^2}\right) t^H \, \mathrm{d}v = t^{-H} \, F\left(\frac{x}{t^H}\right), \tag{4.10}$$

where $F(u) \equiv \int_{-\infty}^{+\infty} G(\sqrt{u^2+v^2})\,\mathrm{d}v$ inherits the normalisation $\int F\,\mathrm{d}u = 1$ from G. The marginal therefore satisfies the same scaling form as the 2D propagator, with the same exponent H but in one dimension. The shape of F distinguishes transport regimes: a Gaussian F characterises ordinary diffusion, while heavy power-law or stretched-exponential tails signal anomalous transport.

The marginal $P(x,t)$ is the object actually estimated from data: use the East component, the North component, or both (if isotropy holds, they coincide and can be pooled; if not, they are analysed separately – see Section 4.2). The scaling collapse consists in plotting $t^H P(x,t)$ against x/t^H for several values of t: all curves should fall on the single function F.

Probability of return to the origin. The value of the propagator at its centre, $P_0(t) \equiv P(\mathbf{0},t)$, i.e. the height of the peak. For a self-similar process it decays as a power law set by the same exponent, $P_0(t) \sim t^{-dH}$ in d dimensions (t^{-2H} in the plane, t^{-H} for a single component). It therefore yields H from the *bulk* of the distribution rather than from its tails, which is valuable when heavy tails make the variance – and hence the MSD – ill-defined; a change of slope in $\log P_0$ versus $\log t$ exposes a crossover between regimes, and a peak exponent that disagrees with the MSD exponent is itself informative, pointing to multi-scaling. (This is the route Mantegna and Stanley used to read off the index of Lévy-like walks.)

Non-Gaussianity. How far the propagator departs from a Gaussian, measured by the non-Gaussian parameter (in the plane)

$$\alpha_2(t) = \frac{\langle |\mathbf{r}(t)|^4 \rangle}{2\,\langle |\mathbf{r}(t)|^2 \rangle^2} - 1, \tag{4.11}$$

which vanishes for a Gaussian and is positive for heavier-than-Gaussian tails (the factor 2 is the two-dimensional Gaussian value of $\langle|\mathbf{r}|^4\rangle/\langle|\mathbf{r}|^2\rangle^2$). Evaluated on the propagator $P(\mathbf{x},t)$ at each elapsed time t, it is a function of t – one number per time – tracing a curve $\alpha_2(t)$ that pinpoints the scales over which the displacement statistics are non-Gaussian (typically largest at intermediate times, where occasional long glides dominate, and decaying if a central-limit Gaussianisation sets in at long times). It summarises the shape of P, it does not replace it.

Velocity autocorrelation $C(\tau)$. $C(\tau) = \langle \mathbf{v}(t_0) \cdot \mathbf{v}(t_0+\tau) \rangle$ (often normalised by $C(0)$) is the dot product of the velocity with its later self: it is large and positive when, a delay τ later, the motion still points the same way, so the scale over which $C(\tau)$ decays *is* the memory time of the heading. It controls the MSD through the Green–Kubo relation (for a stationary velocity) $\mathrm{MSD}(t) = 2\int_0^t (t-\tau)\,C(\tau)\,\mathrm{d}\tau$: a slowly decaying C produces a faster-growing MSD. We do not presume the motion ends in ordinary diffusion; rather, the functional form of $C(\tau)$ is the diagnostic – an integrable C relaxes to normal diffusion, a non-integrable (e.g. power-law) tail sustains persistent super-diffusion with no crossover, and a sign change reveals the anti-persistence expected from circling.

Turning-angle distribution. The distribution of the angle θ between two successive horizontal displacement vectors along the trajectory – the change of heading from one step to the next. A distribution peaked at $\theta = 0$ means strong forward persistence (a correlated random walk); a flat distribution means memoryless reorientation. This is the microscopic origin of the persistence seen in $C(\tau)$ and the MSD.

First-passage and first-exit times. The first-passage time $T(L)$ is the first time the horizontal distance from take-off reaches a value L (the first-exit time, the first time the trajectory leaves a disc of radius L). For a self-similar transport process the characteristic time scales with the length as $\langle T(L) \rangle \sim L^{1/H}$, with dynamic exponent $z = 1/H$, so the slope of $\log\langle T \rangle$ against $\log L$ gives an estimate of H obtained in the space domain, independent of the MSD; the full distribution of $T(L)$ additionally exposes heavy tails coming from long trapping events.

Speed and vertical-velocity distributions. The magnitude of the horizontal velocity and the vertical velocity. Basic kinematic characterization; the vertical velocity also separates climbing from sinking and feeds the segmentation below.

Trajectory geometry.

Tortuosity. How much a path winds, quantified by the ratio between the length actually travelled and the straight-line distance between its endpoints over the same interval (1 for a straight line, larger for a more convoluted path). It is a geometric *efficiency* descriptor and does not map one-to-one onto a transport exponent – very different dynamics can share a tortuosity – so we use it descriptively and to separate phases (close to 1 in glides, large in search and climb) rather than as an estimator of H.

Fractal dimension. The box-counting dimension d_f of the trail, i.e. the exponent with which the number of boxes of size ϵ needed to cover the path grows as $\epsilon \to 0$. For a self-similar walk it is tied to the transport exponent by

$$d_f = \min(1/H,\, 2), \tag{4.12}$$

so a ballistic line ($H = 1$) gives $d_f = 1$, a Brownian path ($H = 1/2$) gives $d_f = 2$, and the reported sub-ballistic soaring ($H \approx 0.88$) gives $d_f \approx 1.14$. The fractal dimension is thus the geometric counterpart of H and a third, independent route to it, after the MSD slope and the return/first-passage scalings; agreement among the three is a strong internal check, disagreement is evidence of multi-scaling.

Radius of gyration. The root-mean-square distance of the trajectory points from their centroid, i.e. the characteristic spatial extent of a flight. As a single number it sets the natural length scale of a flight (useful to normalise other observables); measured as it grows with elapsed time, $R_g(t) \sim t^H$, it offers yet another reading of the same exponent.

One number per flight, or a function of scale? The geometric quantities above can be read in two complementary ways. As a *single number per flight* – one tortuosity, one d_f, one R_g – whose distribution over the ensemble characterises the population and feeds the stratified comparisons. Or as a *function of scale or elapsed time* – $R_g(t)$ as the trajectory unfolds, tortuosity measured on sub-segments of growing duration t, the box count as a function of box size – which is precisely where the transport exponents live ($R_g(t) \sim t^H$; the box-count slope is d_f). The scalar is descriptive; the scale dependence is what links geometry back to the anomalous exponent, and both views will be used.

4.4 Segmentation into flight phases

Building on the previous work of Vilpellet et al. [4], each flight is segmented into the three phases of the soaring cycle:

Transition (glide). The inter-thermal cruise: a near-straight, energy-efficient glide toward the next source of lift (sinking, large directed horizontal displacement). This is the directed "step".

Search. The exploratory phase entered when no thermal is found at a comfortable altitude: gentle turns and probing manoeuvres to sample the air mass and locate the next thermal while limiting altitude loss. It mediates the passage from one step to the next trap.

Climb (thermalling). Circling within rising air to regain altitude: persistent turning with small net horizontal displacement. This is the localized "trap".

That work performs the segmentation with a hidden Markov model whose observations are a small set of *binary* trajectory-derived features — the sign of the vertical speed, the persistence of the curvature angle, and indicators of trajectory straightness.

A first task of this thesis is to scrutinize this segmentation rather than take it for granted: to assess whether those binary variables are adequate and robust on the larger, more heterogeneous dataset assembled here, and whether the feature set or the model should be refined, extended, or replaced. The segmentation method is therefore itself an object of study, and the specific variables are deliberately left open at this stage. This three-state description refines the two-state caricature underlying continuous-time random walks and Lévy walks.

4.5 Phase-resolved observables

The segmentation unlocks a second group of observables, defined per phase occurrence and then as ensemble distributions. These are the building blocks of an intermittent-transport description and, unlike those of Section 4.3, cannot be obtained without first segmenting the flight.

The waiting phase: climb and search. Between two consecutive glides the flier advances little horizontally: it searches for a thermal and climbs in it. For a transport model these two phases together form the *waiting time* between directed steps, so we treat them on an equal footing and build three distributions – the climb duration τ_c, the search duration τ_s, and their sum $\tau_w = \tau_s + \tau_c$, the total time spent essentially stationary between glides. The distinction is not cosmetic: Vilpellet et al. [4] report the search times to be broadly *power-law* distributed while the climb times are closer to *exponential*, so that the heavy tail of the combined waiting time $\psi(\tau_w)$ – the quantity that actually decides whether trapping drives sub-diffusion – would be inherited from the search phase. Whether this still holds on the larger dataset is itself one of the things to re-check once the segmentation is in place. We therefore report $\psi(\tau_w)$ together with the separate $\psi(\tau_c)$ and $\psi(\tau_s)$. Alongside the durations we record the climb kinematics (altitude gained Δz, mean climb rate, thermalling radius and period) and the search geometry (path length, tortuosity, net displacement) – the search being the phase in which prior work locates the signature of pilot skill.

The directed step: glide (transition). Each glide contributes a horizontal step of length ℓ and duration τ_g, with ground speed, glide ratio (horizontal distance per unit altitude lost) and sink rate. Because a glide is close to ballistic, $\ell \approx v_g \, \tau_g$, step length and duration are expected to be nearly proportional; we test this directly through the joint law of (ℓ, τ_g) – the ℓ–τ_g scatter, the conditional mean $\langle \ell \mid \tau_g \rangle$, and the spread of the ratio ℓ/τ_g (the glide speed). Both marginals matter: the tail of the step-length distribution $\lambda(\ell)$ controls super-diffusive contributions, so we compute the distributions of ℓ and τ_g separately as well as their joint law.

Anatomy of a flight. Number of thermals (climbs) per flight, inter-thermal distances, and the fraction of both time and along-track distance spent in each phase: the coarse composition that sets the weight of each phase in the global transport.

Angular diffusion between glides. The turning angle between the heading of one transition segment and the next – the segment-level counterpart of the turning angle of Section 4.3. How fast successive glide directions decorrelate is the key ingredient of the minimal model below.

Two couplings then decide which transport model applies, and are stated here as explicit objects of study. The first is the *step–waiting coupling*: whether a long glide tends to be followed by a long wait, i.e. whether ℓ and τ_w are correlated. A genuine coupling is the defining feature of a Lévy walk and is what separates it from a decoupled CTRW, in which jumps and waiting times are drawn independently.

The second is the *memory in the phase sequence.* A flight is a discrete sequence over the three states $\{\text{transition}, \text{search}, \text{climb}\}$, and memory can hide in two places. (i) The *order* of the phases, summarised by the transition matrix $P_{ij} = \Pr(\text{next} = j \mid \text{current} = i)$: a perfectly cyclic flight $(\mathrm{T} \to \mathrm{S} \to \mathrm{C} \to \mathrm{T})$ gives a near-deterministic matrix, whereas aborted climbs, repeated searches or skipped phases appear as off-cycle probabilities and reveal how stereotyped the soaring cycle actually is. (ii) The *durations*, through correlations between consecutive phase durations – for instance whether a long climb tends to precede a long glide. Both probe the *renewal* assumption underlying the textbook CTRW and Lévy-walk results, in which each step and wait is independent of the history: if the transition matrix is structured or the durations are correlated, the process is non-renewal and those formulas must be amended. Measuring these is therefore what tells us whether a memoryless model is admissible at all.

4.6 Reproducing the reference results

A first objective of the analysis, undertaken once the segmentation is in place, is to check whether the enlarged—and eventually multi-source—dataset reproduces the quantitative and graphical results of Vilpellet et al. [4]. That study covers paragliders, hang gliders and sailplanes; the present data cover the first two (Chapter 3), so both the paraglider and the hang-glider comparisons are within reach, while the sailplane figures become reachable once that source is added. Two figures are the main targets.

The first is the per-phase conditional MSD (their Fig. 3): the transition segments should stay ballistic ($\alpha = 2$, i.e. $H = 1$) across time lags; the search phase should be strongly sub-diffusive ($H \approx 0.3$ for paragliders and hang gliders, ≈ 0.2 for sailplanes); and the climb phase should be quasi-ballistic at long lags but with an effective velocity about two orders of magnitude smaller, including the transient anti-correlation near the $\sim 30\,\mathrm{s}$ thermalling period.

The second is the set of per-phase descriptive statistics (their Fig. 4): the glide ratios (≈ 8.5, 10, and 25 for paragliders, hang gliders, and sailplanes), the horizontal-velocity statistics, the heavy-tailed distribution of transition times, the search-time fraction and the effective search radius, the vertical climb velocities, the thermalling radii, and the thin, near-exponential distribution of climb times—together with the global time budget ($\approx 60\%$ transition, $\approx 35\%$ climb, the remainder searching) and the global sub-ballistic scaling ($H \approx 0.88$).

Recovering these results would validate the acquisition pipeline and the segmentation and would test the universality of the reported behaviour on a substantially larger sample;

systematic departures would be equally informative, pointing to dataset- or method-dependent effects to investigate. This comparison is therefore the bridge between the present data-acquisition stage and the original analysis that follows.

4.7 Transport analysis

Using these observables, test for anomalous transport: estimate the MSD scaling exponent α, fit the tails of $\psi(\tau)$ and $\lambda(\ell)$ with principled methods (e.g. maximum-likelihood power-law fits with goodness-of-fit tests), and perform model selection between the continuous-time random walk and the Lévy-walk frameworks [1, 2] – using the step/waiting-time coupling to tell them apart. Because logging devices, sites, gliders and weather differ, results will be stratified (by season, site, glider class, and data source) and trajectories normalized, and finite-size, censoring and sampling biases will be controlled throughout.

4.8 Modelling and simulation

The aim is a *minimal* stochastic model that reproduces the asymptotic anomalous scaling reported for soaring (a Hurst exponent $H \approx 0.88$) with as few ingredients as possible. A preliminary working hypothesis—to be made precise, and possibly revised, once the empirical phase statistics are available—is that the transport is carried mainly by the transition (glide) phases. The model would therefore resolve the transition segments explicitly while merging search and climb into a single effective non-transition state, so that the soaring cycle reduces to a sequence of directed glides interrupted by reorientations. Anomalous transport would then emerge from the directed transition segments combined with an *angular diffusion* of the heading from one transition to the next: successive glide directions decorrelate gradually rather than abruptly, producing a correlated random walk whose persistence controls the long-time exponent. Simulated ensembles would be compared with the data to test whether these ingredients suffice. These are baseline ideas, to be refined—or changed—as the analysis progresses.

4.9 Tooling

Mirror the acquisition package with an analysis package, and let the figures and tables of this document be generated automatically from the analysis outputs – as the dataset statistics already are – so that this document stays in step with the work and remains a reliable basis for the thesis and for presentations.

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Appendix A

CTRW with power-law waiting times: EA-MSD vs TA-MSD

This appendix derives the ensemble-averaged MSD (EA-MSD) and the time-averaged MSD (TA-MSD) for the continuous-time random walk (CTRW) with heavy-tailed waiting times, and shows explicitly that the two differ and that the TA-MSD depends on the observation time T.

Model

A walker takes steps of length l_i (iid, symmetric, $\langle l^2\rangle = \sigma^2 < \infty$) separated by waiting times τ_i (iid) drawn from

$$\psi(\tau) \sim \frac{\alpha\,\tau_0^\alpha}{\tau^{1+\alpha}}, \qquad \tau \to \infty, \quad 0 < \alpha < 1. \tag{A.1}$$

The mean waiting time diverges ($\langle\tau\rangle = \infty$), so the walker is anomalously slow. Let $n(t)$ be the number of steps completed by time t, $T_n = \sum_{i=1}^{n} \tau_i$ the time of the n-th step, and

$$x(t) = \sum_{i=1}^{n(t)} l_i \tag{A.2}$$

the position at time t.

Ensemble-averaged MSD

Since steps and waiting times are independent,

$$\langle x^2(t)\rangle = \sigma^2\,\langle n(t)\rangle. \tag{A.3}$$

For large t the mean number of renewals in $[0, t]$ follows from the renewal equation and the Tauberian theorem applied to the Laplace transform of ψ. Because $\hat{\psi}(s) \equiv \int_0^\infty e^{-s\tau}\psi(\tau)\,\mathrm{d}\tau \simeq 1 - (\tau_0 s)^\alpha + \cdots$ for $s \to 0$ (a consequence of ψ having a power-law tail with exponent $1+\alpha$), the renewal theorem gives

$$\langle n(t)\rangle \sim \frac{t^\alpha}{\tau_0^\alpha\,\Gamma(1+\alpha)}, \qquad t \to \infty. \tag{A.4}$$

Hence the **EA-MSD** grows sub-linearly,

$$\mathrm{MSD}(t) = \langle x^2(t)\rangle \sim \frac{\sigma^2}{\tau_0^\alpha\,\Gamma(1+\alpha)}\,t^\alpha \equiv K_\alpha\,t^\alpha, \qquad 0 < \alpha < 1. \tag{A.5}$$

The transport is sub-diffusive with Hurst exponent $H = \alpha/2 < 1/2$.

Time-averaged MSD and its mean

For a trajectory of duration T, the time-averaged MSD at lag $t \ll T$ is

$$\overline{\delta^2}(t;T) = \frac{1}{T-t}\int_0^{T-t} \big[x(t_0+t) - x(t_0)\big]^2 \, \mathrm{d}t_0. \tag{A.6}$$

Since $\langle [x(t_0+t) - x(t_0)]^2 \rangle = \sigma^2 \langle n(t_0+t) - n(t_0) \rangle$, the mean of the integrand equals σ^2 times the mean number of steps in any window of width t. The key observation is that the integral $\int_0^{T-t} [n(t_0+t) - n(t_0)]\, \mathrm{d}t_0$ counts, for each step k that lands at time $T_k \in [0,T]$, the length of the t-wide sliding window that contains T_k. For T_k deep in the interior ($t \le T_k \le T-t$) this length is exactly t; the contributions from the two boundary layers of width t are negligible for $t \ll T$. Therefore

$$\int_0^{T-t} [n(t_0+t) - n(t_0)]\, \mathrm{d}t_0 \approx t\, n(T), \qquad t \ll T, \tag{A.7}$$

and

$$\overline{\delta^2}(t;T) \approx \frac{\sigma^2\, t\, n(T)}{T}. \tag{A.8}$$

Taking the ensemble average and using $\langle n(T) \rangle \sim K_\alpha T^\alpha / \sigma^2$,

$$\mathrm{TAMSD}(t,T) \equiv \langle \overline{\delta^2}(t;T) \rangle \approx \sigma^2 \, \frac{\langle n(T) \rangle}{T}\, t \sim K_\alpha \, T^{\alpha-1}\, t. \tag{A.9}$$

Comparing (A.5) and (A.9):

- The EA-MSD scales as t^α (sub-diffusive), whereas the TAMSD scales as t^1 (linear in the lag) – a qualitatively different dependence on t.
- The TAMSD carries an explicit $T^{\alpha-1}$ prefactor: since $\alpha < 1$ this is a decreasing function of T, so longer flights yield smaller TAMSD at the same lag. This T-dependence is the signature of *aging*: the process has not forgotten its past, and the answer depends on how long we have observed it.
- The ratio $\mathrm{TAMSD}(t,T)/\mathrm{MSD}(t) \sim (t/T)^{1-\alpha} \to 0$ as $T \to \infty$ at fixed t – the two observables diverge.

Distribution of the TA-MSD: non-self-averaging

Equation (A.8) shows that $\overline{\delta^2}(t;T) \approx \sigma^2 t\, n(T)/T$, so the randomness in the TA-MSD comes entirely from $n(T)$, the total number of steps in $[0,T]$. For a power-law renewal process the rescaled count

$$\xi \equiv \frac{n(T)}{\langle n(T) \rangle} \tag{A.10}$$

converges in distribution as $T \to \infty$ to a non-degenerate limit law: the *Mittag-Leffler distribution* M_α, whose probability density function is defined through its moments $\langle \xi^n \rangle = n!\,\Gamma(1+\alpha)^n / \Gamma(1+n\alpha)$. This is a broad, non-Gaussian distribution on $(0,\infty)$ that does *not* concentrate around $\xi = 1$ for $\alpha < 1$: different trajectories give systematically different values of $\overline{\delta^2}(t;T)$ even in the limit $T \to \infty$. This is *non-self-averaging*: the TA-MSD of a single trajectory is not a reliable estimate of any ensemble quantity, regardless of how long the trajectory is.

Quantitatively, the ergodicity-breaking parameter

$$\mathrm{EB}(t,T) \equiv \frac{\langle [\overline{\delta^2}(t;T)]^2 \rangle - \langle \overline{\delta^2}(t;T) \rangle^2}{\langle \overline{\delta^2}(t;T) \rangle^2} = \mathrm{Var}(\xi) \sim c_\alpha \, T^{-\alpha} \to 0 \tag{A.11}$$

does tend to zero (the variance of ξ decreases with T), but the *distribution* of ξ converges to a non-trivial limit rather than to a delta function. This is why monitoring the full distribution of $\overline{\delta^2}^{(k)}(t)$ across trajectories – not just its mean – is the proper diagnostic for non-ergodicity: the collapse (or non-collapse) of that distribution as T grows tells us whether we are dealing with a self-averaging ergodic process or a non-self-averaging non-ergodic one.

For comparison, an ordinary random walk ($\alpha = 1$, finite mean waiting time) has $n(T)/\langle n(T)\rangle \to 1$ in probability as $T \to \infty$, so the distribution of the TA-MSD collapses to a delta function, $\overline{\delta^2}(t;T) \to \mathrm{MSD}(t)$ almost surely, and the process is ergodic.

Appendix B

Power spectral density and Welch's method

This appendix explains the tool used in Chapter 4 to compare the two altitude channels of an IGC file, namely the *power spectral density* (PSD) estimated with *Welch's method*. It is written so the reader need not already know what these are.

What the power spectral density is

Take a signal sampled at a fixed rate, here an altitude series $z_n = z(n\,\Delta t)$, $n = 0, \ldots, N-1$, with sampling interval Δt (so the sampling frequency is $f_s = 1/\Delta t$). Any such finite signal can be written as a sum of sinusoids of different frequencies (its discrete Fourier transform). The **power spectral density** answers a simple question: *how is the "power" (the variance) of the signal distributed across frequencies?* A slow drift puts its power at low frequency; rapid jitter puts its power at high frequency. Formally the PSD $S(f)$ is the Fourier transform of the autocovariance of the signal, and it is normalised so that the total area under it equals the variance of the signal,

$$\int_0^{f_s/2} S(f)\,\mathrm{d}f \;=\; \mathrm{Var}(z), \tag{B.1}$$

with units of (metres)^2 per hertz. The upper limit $f_s/2$ is the *Nyquist frequency*: a series sampled every Δt cannot resolve anything faster than one oscillation per two samples. Reading a PSD is therefore reading *where in frequency* the signal lives: this is exactly what separates a slowly drifting but locally clean channel (power concentrated at low f) from a channel with a high-frequency noise floor (power that stays large up to the Nyquist frequency).

The periodogram and why it is noisy

The naive estimator of $S(f)$ is the **periodogram**: take the discrete Fourier transform $\hat{z}(f) = \sum_n z_n\, e^{-2\pi i f n \Delta t}$ and form $|\hat{z}(f)|^2$ (suitably normalised). It is unbiased but *not consistent*: its variance does not shrink as the record gets longer – a longer signal gives more frequency points, each still as noisy as before. Plotted directly it is a spiky, unreadable curve, from which one cannot reliably compare two channels.

Welch's method: segment, window, average

Welch's method turns the noisy periodogram into a usable estimate by *averaging*. The recipe is:

(i) **Segment.** Split the record into several shorter, overlapping segments (here of length $N_{\text{seg}} = 256$ samples, the standard 50% overlap).

(ii) **Window.** Multiply each segment by a smooth taper (a Hann window) that falls to zero at its ends. This suppresses the spurious high-frequency power (*spectral leakage*) that an abrupt segment boundary would otherwise inject.

(iii) **Average.** Compute the periodogram of each windowed segment and average them.

Averaging K segments cuts the variance of the estimate by roughly a factor K, at the cost of coarser frequency resolution (set by the segment length rather than the whole record). The result is a smooth PSD estimate whose *level* can be compared meaningfully between two signals. This is the standard, robust spectral estimator; here it is provided by `scipy.signal.welch`. Before the transform each segment is *linearly detrended*, which removes the mean and the overall climb/drift ramp so that the estimate reflects the fluctuations about the trend – the part where sensor noise lives – rather than the trend itself.

Computational details for the IGC dataset

IGC loggers record one fix per second nominally, but the timestamps are not guaranteed to be perfectly regular: small gaps (a missed fix, a logger that writes at 1.05 s instead of 1.00 s) are common. Welch's method requires a *uniformly sampled* signal. The following procedure is applied to every flight before the transform.

(i) **Modal sampling period.** For each flight the median inter-fix interval is computed and rounded to the nearest integer second. The value that appears most often across all flights in the subsample – the *mode* – is taken as the common target Δt (in practice $\Delta t = 1\,\text{s}$ for both disciplines, giving $f_s = 1\,\text{Hz}$ and $f_{\text{Nyq}} = 0.5\,\text{Hz}$).

(ii) **Flight selection.** Only flights whose own median period lies within 0.25 s of the modal Δt are retained, so that the pooled spectra all share one frequency axis. Flights with fewer than $N_{\text{seg}} = 256$ fixes (roughly 4 minutes at $\Delta t = 1\,\text{s}$) are also excluded, since at least one full segment is needed. Of the 400 sampled flights per discipline (with a barometric channel and enough fixes), this leaves 212 paragliders but only 114 hang gliders: paragliders are overwhelmingly 1 s loggers, so the population mode is a good match for most of them, whereas hang gliders split across several native rates (1, 2, 5, ≥ 8 s, Figure 4.7) and only their 1 s-logger minority survives this cut. The hang-glider PSD below is therefore an average over that minority, not the full hang-glider population – an honest reading of the panel, not a claim about every hang-glider logger.

(iii) **Uniform resampling.** The altitude series $z(t)$ is linearly interpolated onto a grid $t_0, t_0 + \Delta t, t_0 + 2\,\Delta t, \ldots$ spanning the flight. This replaces the mildly irregular original timestamps with a perfectly uniform series at the modal cadence, without introducing frequency content beyond f_{Nyq}.

(iv) **Welch on the resampled series.** `scipy.signal.welch` is called with $f_s = 1/\Delta t$, $N_{\text{seg}} = 256$, 50 % overlap, a Hann window, and linear detrending per segment (as described above). The resulting f-axis runs from 0 to $f_{\text{Nyq}} = f_s/2$, the last resolvable frequency.

(v) **Ensemble average.** The PSD of each qualifying flight is added to a running sum, separately for each discipline and channel (barometric / GNSS). The mean over all contributing flights is the curve shown in Figure 4.1(c).

Because the frequency axis is determined entirely by Δt – which is the same for every retained flight by construction – the dotted line in the figure marking $f_{\mathrm{Nyq}} = 1/(2\,\Delta t)$ is a single fixed value, not a flight-by-flight average.

Reading the altitude spectra

In Figure 4.1(c) the PSD is computed this way for the barometric and the GNSS altitude of each flight and averaged over the sampled ensemble, at the common sampling cadence. Two features carry the argument of Chapter 4. First, both channels agree at *low* frequency, where the real vertical motion (climbs and glides) lives. Second, towards the *Nyquist* frequency the GNSS curve flattens into a high plateau – a white-ish noise floor – lying about two to three orders of magnitude above the barometric curve, which keeps falling. That gap is the high-frequency GNSS jitter, precisely the band that a time derivative (vertical velocity) or a displacement increment amplifies, and it is the quantitative reason the vertical dynamics are taken from the barometric channel.